ORACLE

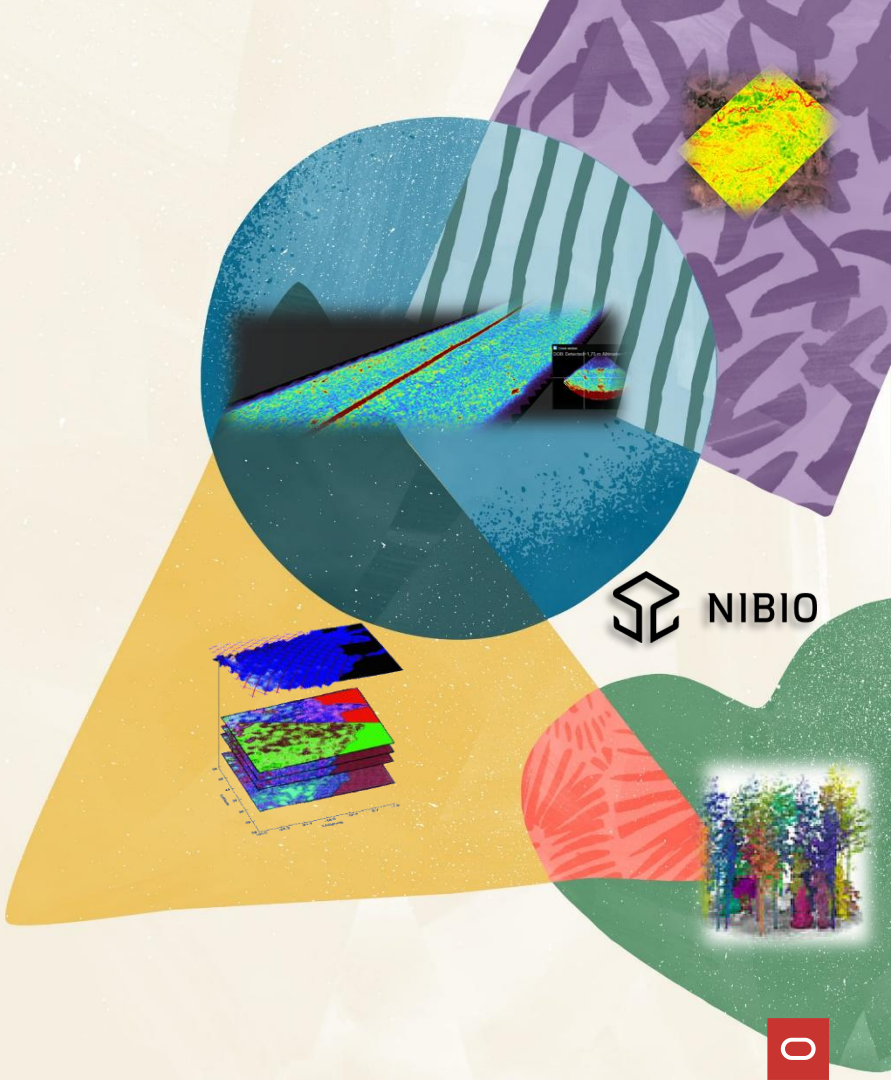
Oracle Spatial

Ai4Nature

Maryam Sajjadian

Spatial & Graph

Black Belt, EMEA

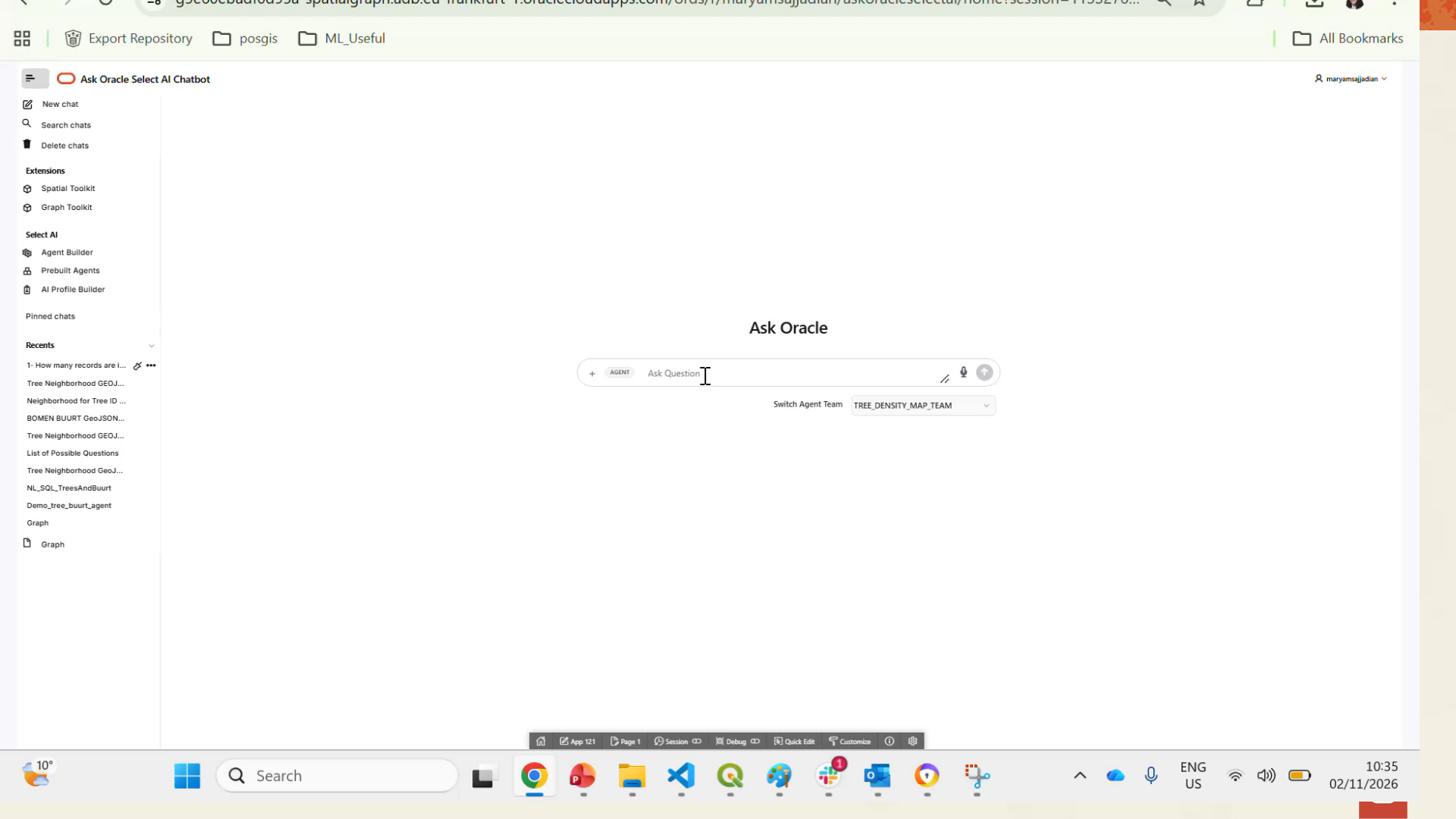


SMART Forest

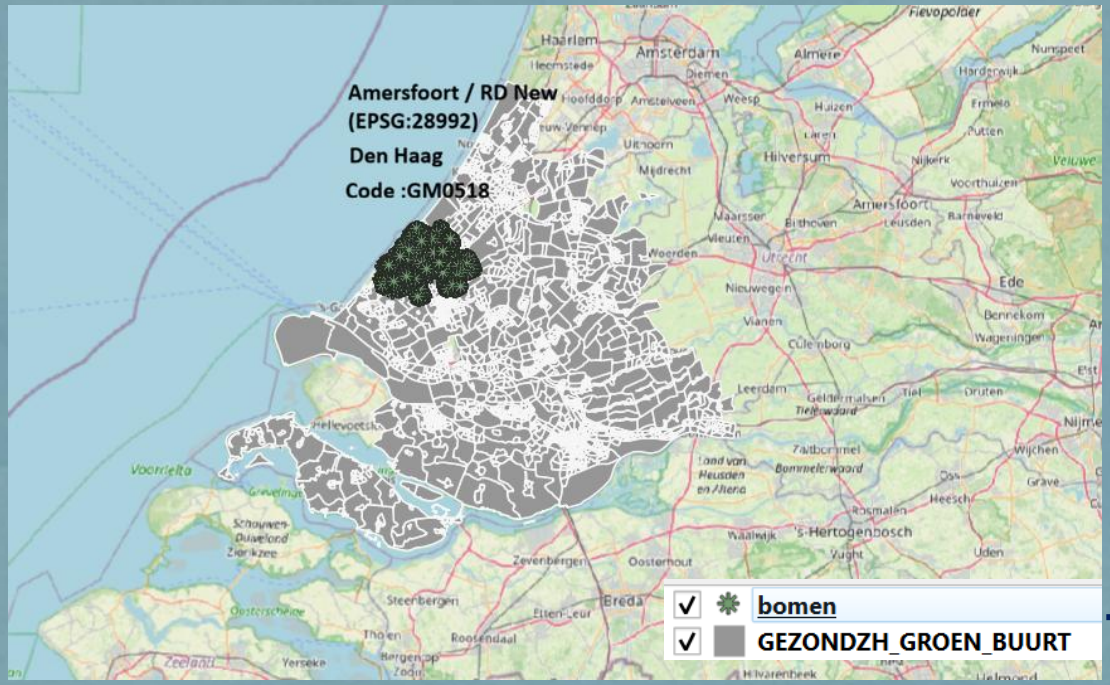
- Introduction
- Datasets
- Toolsets and components
- Demo



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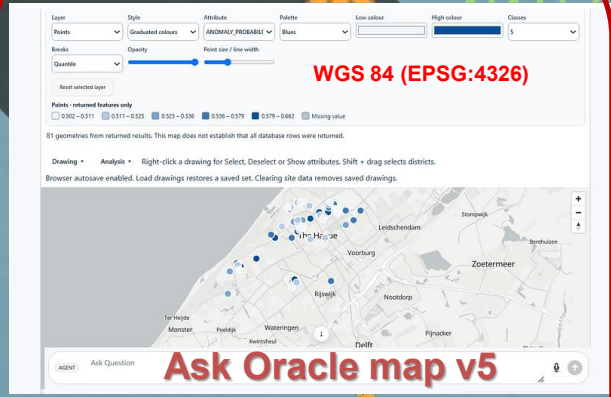

Export Repository
posgis
ML_Useful
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Ask Oracle Select AI Chatbot
maryamsajjadian
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Search chats
Delete chats
Extensions
Spatial Toolkit
Graph Toolkit
Select AI
Agent Builder
Prebuilt Agents
AI Profile Builder
Pinned chats
Recents
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Neighborhood for Tree ID ...
BOMEN BUURT GeoJSON...
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List of Possible Questions
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Datasets



Source layers

Spatial Studio



BOMEN_TREES
GEZOND_GREEN_NEIGHBOURHOOD

Oracle tables

1- Greeting
2- SQL, functions , procedures
3- OML algorithms in database
4- NL2SQL and workflow
5- History of questions

Tools & Components

26ai

- Select AI (NL2SQL, RAG, Agent)

```
BEGIN
 DBMS_CLOUD_AI.CREATE_PROFILE(
   profile_name => 'GENAI',
   attributes => '{"provider": "oci",
    "credential_name": "GENAI_CRED"
    "object_list": [{"owner": "SH", "name": "customers"},
                    {"owner": "SH", "name": "countries"},
                    {"owner": "SH", "name": "supplementary_demographics"}
                    {"owner": "SH", "name": "profits"},
                    {"owner": "SH", "name": "promotions"},
                    {"owner": "SH", "name": "products"}],
     "oci_compartment_id": "ocid1.compartment.oc1...",
              "oci_endpoint_id": "ocid1.generativeaiendpoint.oc1.us-chicago-1....",
              "region": "us-chicago-1",
              "model": "cohere.command-light",
              "oci_apiformat": "COHERE"
    }');
END;
/
```

Profile Name

Provider

Credentials

Object List

Tools & Components



- Select AI (NL2SQL, RAG, Agent)

Create an AI profile → DBMS_CLOUD_AI.create_profile

Create the vector index → DBMS_CLOUD_AI.create_vector_index

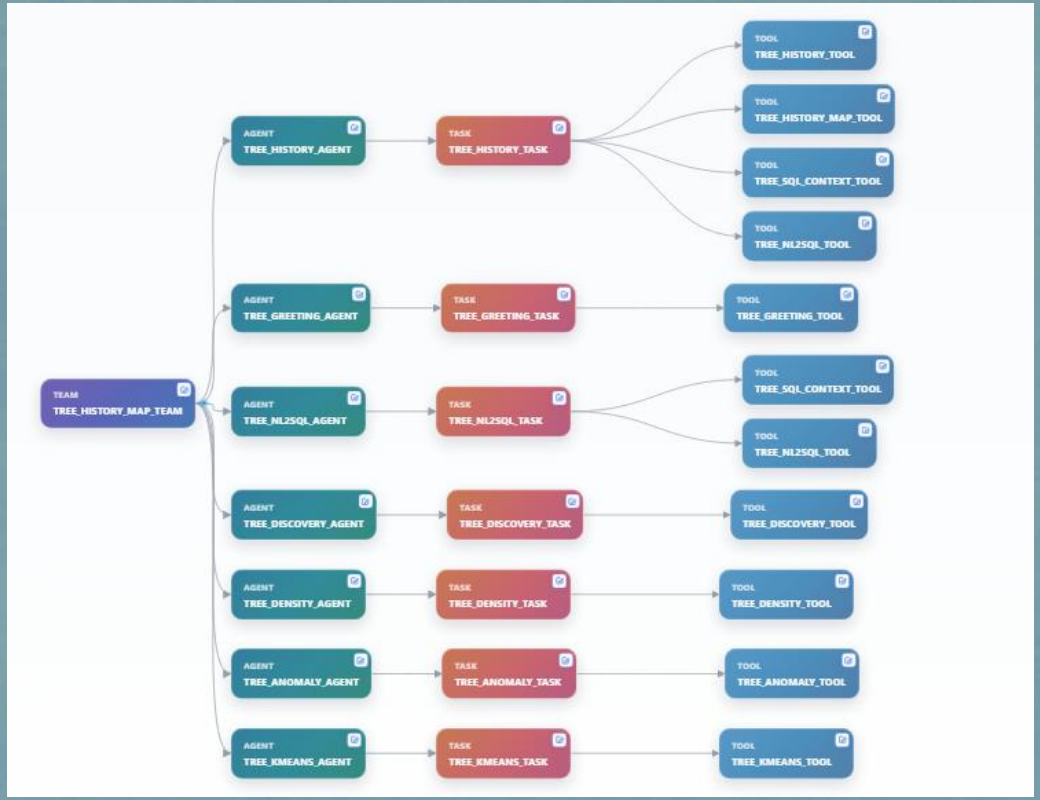
Start asking questions...

```
SQL> SELECT AI NARRATE <prompt>
or
DBMS_CLOUD_AI.generate (<prompt>, <AI profile>, 'narrate');
```

Tools & Components

26ai

- Select AI (NL2SQL, RAG, Agent)



Tools & Components



- **OML Algorithm**

- **OML K-means** — groups trees by age and trunk-diameter class.
- **OML one-class SVM** — flags unusual age and diameter combinations for review.

```
BEGIN
  EXECUTE IMMEDIATE 'DROP TABLE cluster_settings';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/

CREATE TABLE cluster_settings (
  setting_name  VARCHAR2(30),
  setting_value VARCHAR2(4000)
);

BEGIN
  INSERT INTO cluster_settings VALUES (DBMS_DATA_MINING.ALGO_NAME, DBMS_DATA_MINING.ALGO_KMEANS);
  INSERT INTO cluster_settings VALUES (DBMS_DATA_MINING.CLUS_NUM_CLUSTERS, '3');
  COMMIT;
END;
```

```
BEGIN
    DBMS_DATA_MINING.CREATE_MODEL(
        model_name          => 'TREE_ATTRIBUTES_CLUSTER_MODEL',
        mining_function     => DBMS_DATA_MINING.CLUSTERING,
        data_table_name     => 'BOMEN_OML_V',
        case_id_column_name => 'ID',
        settings_table_name => 'CLUSTER_SETTINGS'
    );
END;
```

Tools & Components



- **AI Vector Search** — finds similar saved questions and answers using the ONNX embeddings.

ONNX (`ALL_MINILM_L12_V2`) — embeddings for history search.

Use a database resident embedding model

Generate embeddings using the VECTOR_EMBEDDING() SQL function using an imported ONNX embedding model so that no data leaves the database

```
-- Load embedding model
BEGIN
   dbms_vector.load_onnx_model (
   directory => 'DATA_DUMP_DIR',
   file_name => 'all_MiniLM_L12_v2.onnx',
   model_name => 'allMiniLML12',
   metadata => JSON('{"function" : "embedding", "embeddingOutput" : "embedding",
                      "input": {"input": ["DATA"]}}')
   );
END ;
```

There is currently a size limit on the size of the onnx file (Open Neural Network Exchange format.)

Questions



- **Oracle Spatial** — stores tree points and neighbourhood polygons; calculates density and converts results to GeoJSON.
  - Show 11 trees closest to tree ID 5811455, including that tree.
  - Which neighbourhood contains tree ID 5811455?
  - Which neighbourhood has the highest tree density?
  - How are *Quercus robur* trees grouped by age and trunk-diameter class?

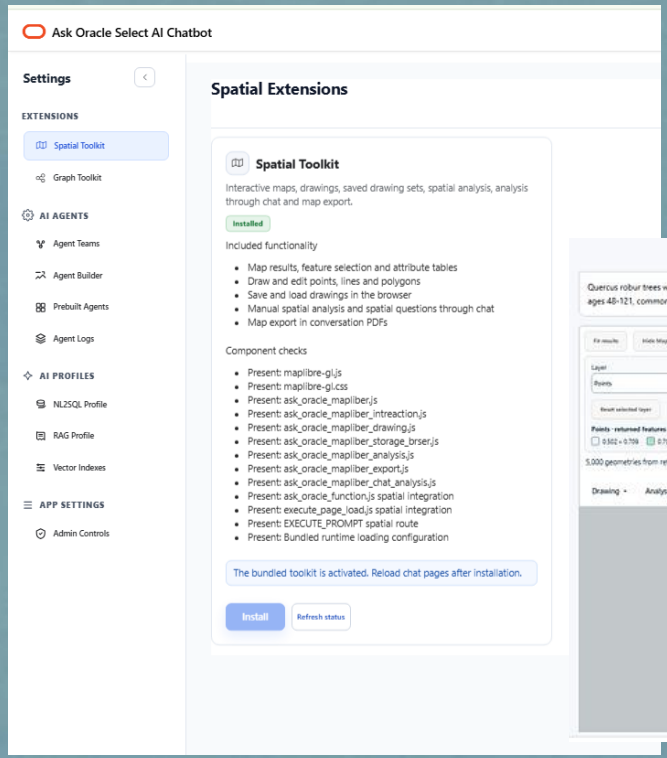


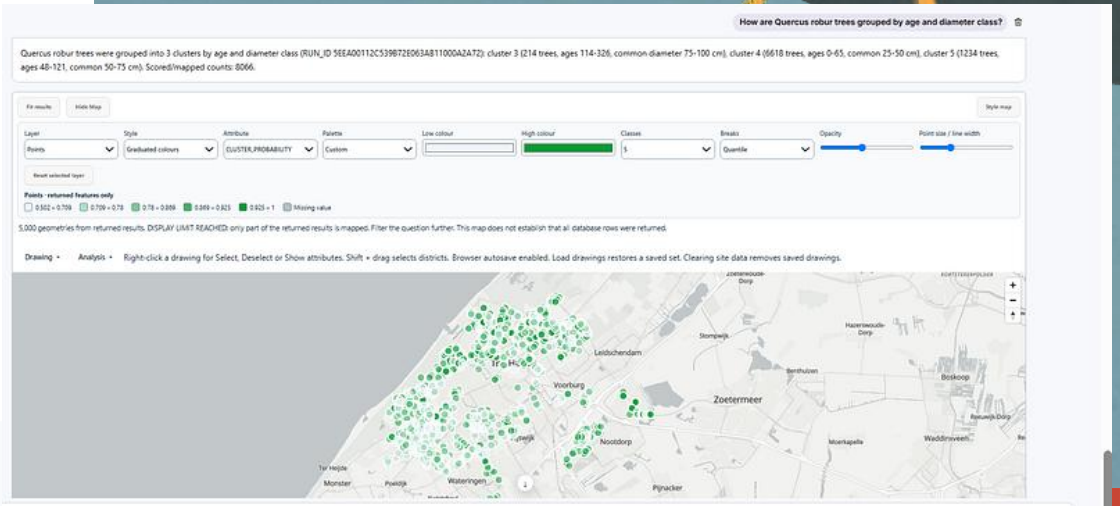
- **Nonspatial**
  - Hi—what questions can you answer?
  - Which tree species are most common?
  - How many trees are recorded in each neighbourhood?
  - Which neighbourhoods have less than 30% green area?
  - What did I ask earlier about K-means, and what was the answer?

Oracle APEX

26ai

- **Ask Oracle v5 (APEX + MapLibre)** — chat interface and interactive map.

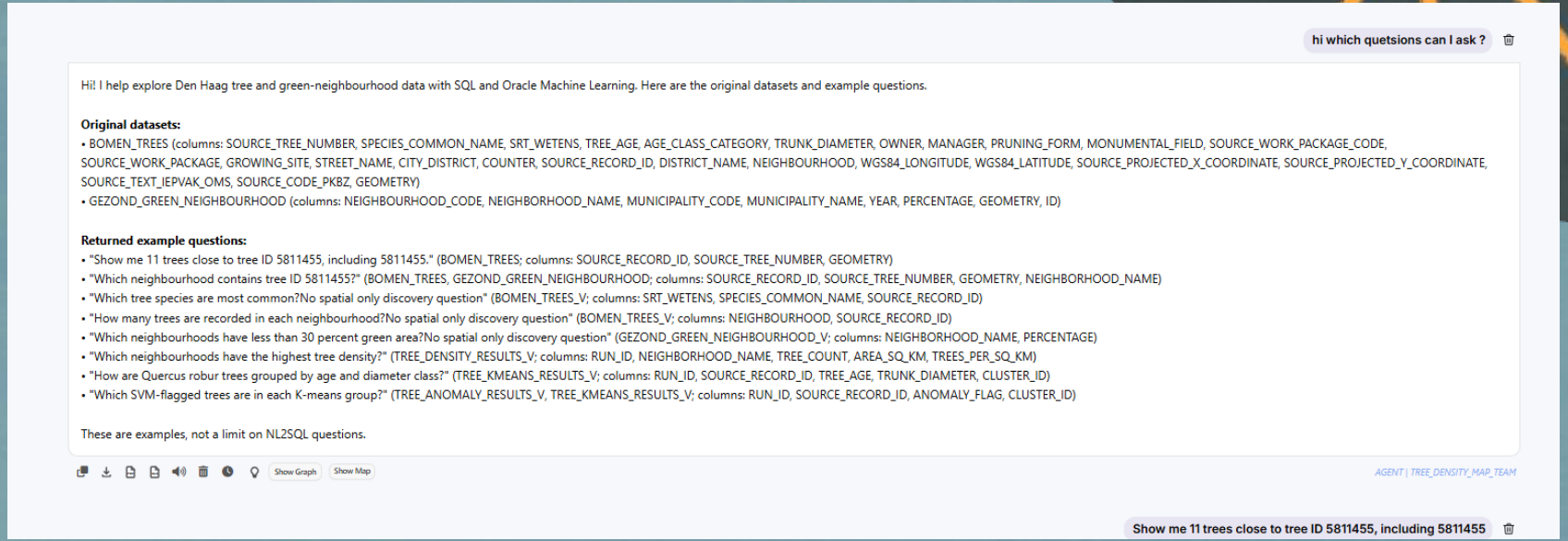




Oracle APEX

26ai

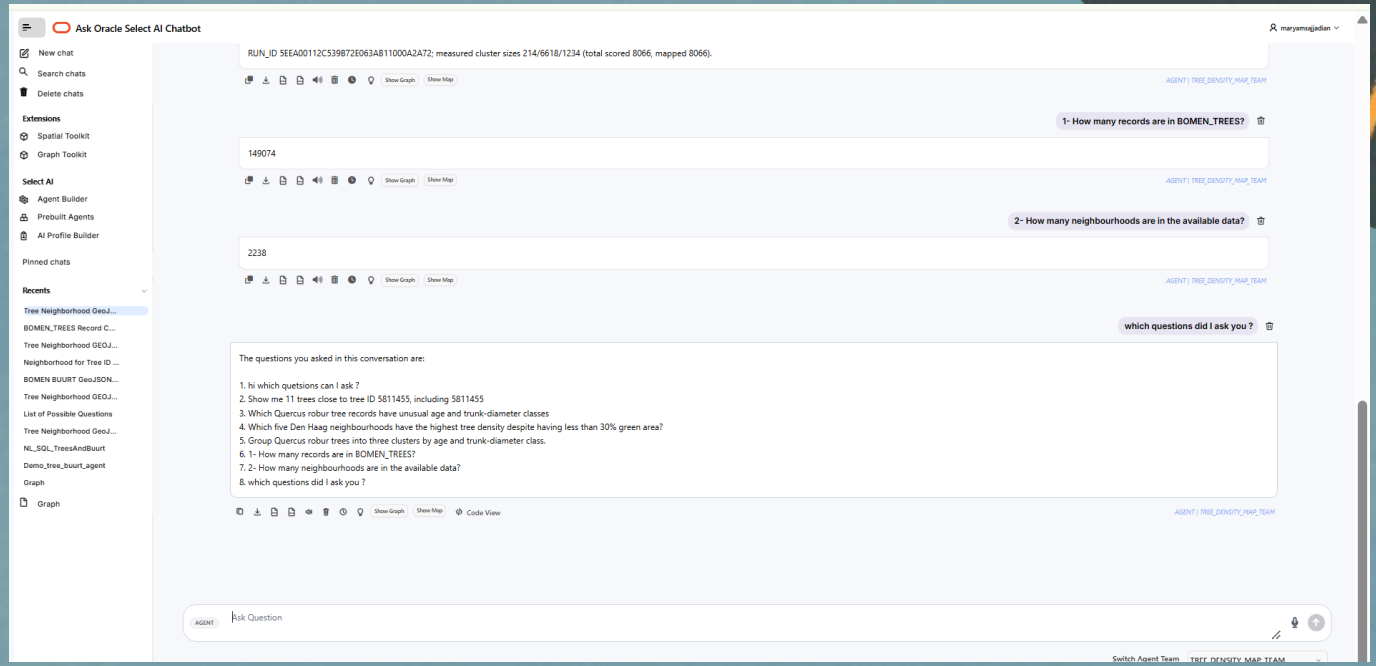
- **Ask Oracle v5 (APEX + MapLibre)** — chat interface and interactive map.

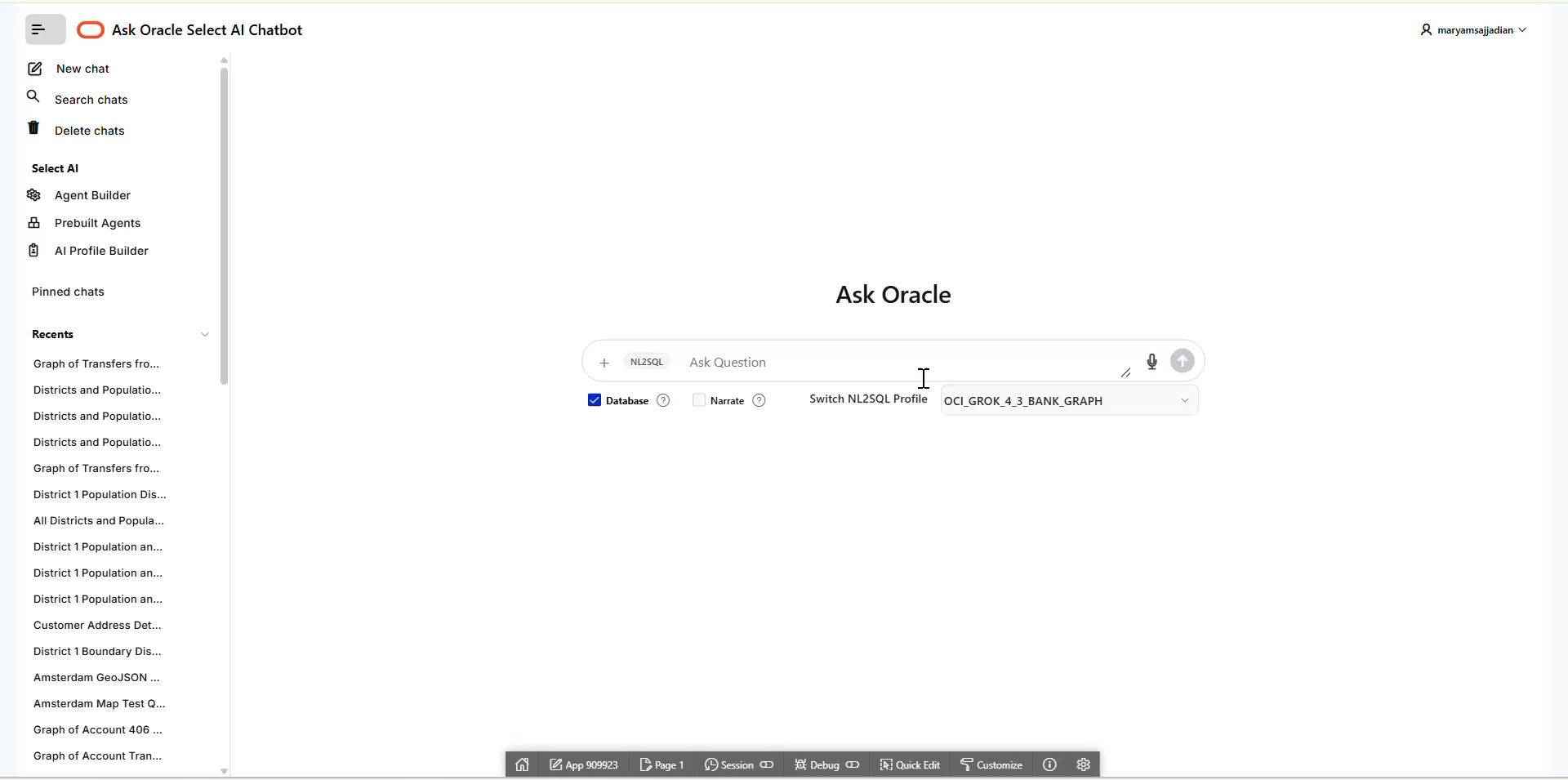


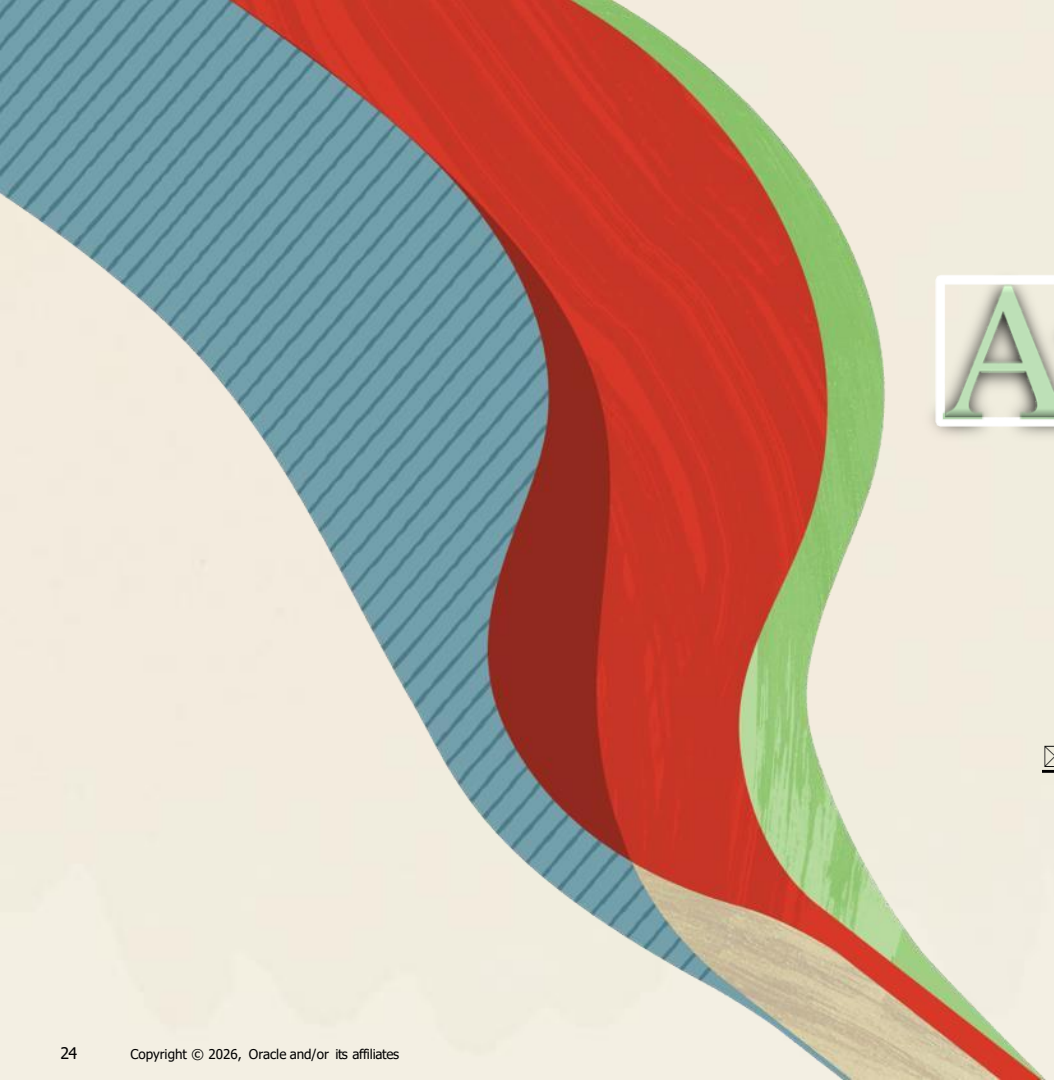
Oracle APEX

26ai

- **Ask Oracle v5 (APEX + MapLibre)** — chat interface and interactive map.




Ask Oracle Select AI Chatbot
maryamsajjadian
New chat
Search chats
Delete chats
Select AI
Agent Builder
Prebuilt Agents
AI Profile Builder
Pinned chats
Recents
Graph of Transfers fro...
Districts and Populatio...
Districts and Populatio...
Districts and Populatio...
Graph of Transfers fro...
District 1 Population Dis...
All Districts and Popula...
District 1 Population an...
District 1 Population an...
District 1 Population an...
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District 1 Boundary Dis...
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App 909923
Page 1
Session
Debug
Quick Edit
Customize



NIBIO

AI4Nature

Thank you

✉ Maryam.sajjadian@oracle.com

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